Zihao Zhu

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Google Scholar (1,000+ citations)

Education

The Chinese University of Hong Kong, Shenzhen — **Shenzhen, China**
Ph.D. in Data Science, School of Data Science — *2021–2026*
Advisor: Baoyuan Wu

University of Chinese Academy of Sciences — **Beijing, China**
M.S. in Cybersecurity, Institute of Information Engineering — *2018–2021*
Advisor: Jing Yu

China University of Mining and Technology — **Xuzhou, China**
B.Eng. in Computer Science and Technology, School of Computer Science and Technology — *2014–2018*

Research Interests

Trustworthy AI; safety and alignment of large language and reasoning models; risk-aware embodied agents; multimodal data quality and content safety.

Research Experience

University at Buffalo, State University of New York — **Buffalo, NY, USA**
Incoming Postdoctoral Researcher, Pending visa approval — *Start date TBD*

Huawei 2012 Shield Lab — **Shenzhen, China**
Research Collaborator, University–Industry Collaboration — *Dec. 2024–Oct. 2025*

- Identified and systematically studied the "unthinking vulnerability" in reasoning models; developed Breaking of Thought (BoT) attacks and the plug-and-play Monitoring of Thought (MoT) framework to improve reasoning efficiency and safety. (NeurIPS 2025 Workshop)
- Developed AdvChain, an adversarial chain-of-thought tuning paradigm that teaches models to self-correct harmful reasoning drift, improving jailbreak robustness while reducing over-refusal. (ICLR 2026)

Tencent AI Lab — **Shenzhen, China**
Research Intern — *Mar. 2024–Jul. 2024*

- Led the development of EARBench, the first benchmark for evaluating physical-risk awareness in embodied-AI task planning; built EARDataset through multi-agent generation of safety-critical scenarios.

Publications

20 published or accepted journal, conference, and workshop papers (8 as first author), plus 4 preprints (3 as first author). Author name is shown in **bold**.

(TPAMI: 3; IJCV: 2; Neurocomputing: 1; NeurIPS: 4; ICLR: 3; CVPR: 1; IJCAI: 2; AAAI: 1; ICASSP: 2; PR: 1)

Large Language Model Safety and Reasoning

1. FVH-Bench: Evaluating Image Generator Safety Beyond Overtly Harmful Imagery
   Zihao Zhu, *Yuhan Liu, Siwei Lyu, Baoyuan Wu* (arXiv 2026; Submitted to ICLR 2027)
2. To Think or Not to Think: Exploring the Unthinking Vulnerability in Large Reasoning Models
   Zihao Zhu, *Hongbao Zhang, Ruotong Wang, Xu Ke, Siwei Lyu, Baoyuan Wu* (NeurIPS 2025 Workshop)
3. Loong: Synthesize Long Chain-of-Thoughts at Scale through Verifiers
   Project participant: ***Zihao Zhu*** (NeurIPS 2025 Workshop)
4. AdvChain: Adversarial Chain-of-Thought Tuning for Robust Safety Alignment of Large Reasoning Models
   Zihao Zhu, *Xinyu Wu, Gehan Hu, Siwei Lyu, Ke Xu, Baoyuan Wu* (ICLR 2026)

AI Agents

5. Stealth Apart, Harm Together: Skill Cascading Attacks on Skill-Based Agent Systems
   Zihao Zhu, *Siwei Lyu, Adel Bibi, Baoyuan Wu* (NeurIPS 2026)
6. EARBench: Towards Evaluating Physical Risk Awareness for Task Planning of Foundation Model-based Embodied AI Agents
   Zihao Zhu, *Bingzhe Wu, Zhengyou Zhang, Lei Han, Qingshan Liu, Baoyuan Wu* (arXiv 2024)
7. BrandFusion: A Multi-Agent Framework for Seamless Brand Integration in Text-to-Video Generation
   Zihao Zhu, *Ruotong Wang, Siwei Lyu, Min Zhang, Baoyuan Wu* (CVPR 2026 Findings)

Data Governance

8. VDC: Versatile Data Cleanser based on Visual-Linguistic Inconsistency by Multimodal Large Language Models
   Zihao Zhu, *Mingda Zhang, Shaokui Wei, Bingzhe Wu, Baoyuan Wu* (ICLR 2024)
9. HMGIE: Hierarchical and Multi-Grained Inconsistency Evaluation for Vision-Language Data Cleansing
   Zihao Zhu, *Hongbao Zhang, Guanzong Wu, Siwei Lyu, Baoyuan Wu* (arXiv 2024)

AI Security

10. Attacks in Adversarial Machine Learning: A Systematic Survey from the Life-cycle Perspective
    Baoyuan Wu, ***Zihao Zhu***, *Li Liu, Qingshan Liu, Zhaofeng He, Siwei Lyu* (IJCV 2025)
11. Defenses in Adversarial Machine Learning: A Survey
    Baoyuan Wu, Shaokui Wei, Mingli Zhu, Meixi Zheng, ***Zihao Zhu***, *Mingda Zhang, Hongrui Chen, Danni Yuan, Li Liu, Qingshan Liu* (IEEE TPAMI 2025)
12. BlackboxBench: A Comprehensive Benchmark of Black-box Adversarial Attacks
    Meixi Zheng, Xuanchen Yan, ***Zihao Zhu***, *Hongrui Chen, Baoyuan Wu* (IEEE TPAMI 2025)
13. BackdoorBench: A Comprehensive Benchmark and Analysis of Backdoor Learning
    Baoyuan Wu, Hongrui Chen, Mingda Zhang, ***Zihao Zhu***, *Shaokui Wei, Danni Yuan, Mingli Zhu, Ruotong Wang, Li Liu, Chao Shen* (IJCV 2025)
14. Versatile Backdoor Attack with Visible, Semantic, Sample-Specific, and Compatible Triggers
    Ruotong Wang, Hongrui Chen, ***Zihao Zhu***, *Li Liu, Baoyuan Wu* (IEEE TPAMI 2024)
15. Reliable Poisoned Sample Detection against Backdoor Attacks Enhanced by Sharpness Aware Minimization
    Mingda Zhang, Mingli Zhu, ***Zihao Zhu***, *Li Shen, Baoyuan Wu* (ICLR 2026)
16. BackdoorBench: A Comprehensive Benchmark of Backdoor Learning
    Baoyuan Wu, Hongrui Chen, Mingda Zhang, ***Zihao Zhu***, *Shaokui Wei, Danni Yuan, Chao Shen* (NeurIPS 2022)
17. Boosting Backdoor Attack with a Learnable Poisoning Sample Selection Strategy
    Zihao Zhu, *Mingda Zhang, Shaokui Wei, Li Shen, Yanbo Fan, Baoyuan Wu* (Neurocomputing 2026)

Multimodal Learning

18. From Shallow to Deep: Compositional Reasoning over Graphs for Visual Question Answering
    Zihao Zhu (ICASSP 2022)
19. Mucko: Multi-Layer Cross-Modal Knowledge Reasoning for Fact-based Visual Question Answering
    Zihao Zhu, *Jing Yu, Yujing Wang, Yajing Sun, Yue Hu, Qi Wu* (IJCAI 2020)
20. Cross-Modal Knowledge Reasoning for Knowledge-based Visual Question Answering
    Jing Yu, ***Zihao Zhu***, *Yujing Wang, Weifeng Zhang, Yue Hu, Jianlong Tan* (Pattern Recognition 2020)

Other Publications

21. Learning to Optimize Permutation Flow Shop Scheduling via Graph-based Imitation Learning
    Longkang Li, Siyuan Liang, ***Zihao Zhu***, *Xiaochun Cao, Chris Ding, Hongyuan Zha, Baoyuan Wu* (AAAI 2024)
22. Enhanced Few-Shot Class-Incremental Learning via Ensemble Models
    Mingli Zhu, ***Zihao Zhu***, *Sihong Chen, Chen Chen, Baoyuan Wu* (arXiv 2024)
23. MCR-NET: A Multi-Step Co-Interactive Relation Network for Unanswerable Questions on Machine Reading Comprehension
    Wei Peng, Yue Hu, Luxi Xing, Yuqiang Xie, ***Zihao Zhu***, *Yajing Sun* (ICASSP 2021)
24. DAM: Deliberation, Abandon and Memory Networks for Generating Detailed and Non-repetitive Responses
    Xiaoze Jiang, Jing Yu, Zengchang Qin, ***Zihao Zhu***, *Qi Wu* (IJCAI 2020)

Academic Service

Reviewer: IEEE Transactions on Image Processing; ICLR; ICML; NeurIPS; CVPR; ICCV; AAAI; AISTATS

Awards and Honors

2024: Guotai Junan Scholarship
2021 AIRS Ph.D. Research Talent Program
2020: Merit Student, University of Chinese Academy of Sciences
2018 Outstanding Undergraduate Thesis Award
2015, 2016: First-Prize Scholarship, China University of Mining and Technology